

Assignment Questions

Okay, I understand! Since you haven't provided me with any faculty notes, textbooks, or reference books related to Java, I will have to create the assignment questions based purely on my existing knowledge and common Java concepts. I'll aim for a variety of question types (short, long, and application-based) as requested.

Important Note: *Because I lack specific faculty notes, these questions are designed to be fairly generic and cover fundamental Java topics. A real assignment would rely heavily on the information provided in your course materials.*

Here are 5 Assignment Questions for a Java Course:

Assignment Questions - Java

1. Short Question (Conceptual)

Question: Explain the difference between `==` and `.equals()` when comparing String objects in Java. Why is this distinction important?

Expected Response Length: Approximately 3-5 sentences.

(Concept Covered): Object Equality, Strings

2. Long Question (Code Understanding & Explanation)

Question: Consider the following Java code snippet:

```
```java
```

```
public class Example {
```

```
 public static void main(String[] args) {
```

```
 int x = 10;
```

```
int y = 5;

int sum = x + y;

System.out.println("The sum is: " + sum);

}

}

...

```

\* Write a paragraph explaining what the code does, including the purpose of each line and the expected output. Explain the concept of variable scope in this example.

\* **(Concept Covered):** Basic Java Syntax, Variables, Operators, Output Statements, Scope

### **3. Application Based Question (Problem Solving)**

\* **Question:** You need to write a Java program that calculates the factorial of a non-negative integer entered by the user. Write pseudocode *and then* provide a short Java code snippet to implement this functionality. Consider error handling if the user enters invalid input (e.g., negative numbers or characters).

\* **Expected Response Length:** Pseudocode - 3-5 lines; Code Snippet – 10-20 Lines of Java Code.

\* **(Concept Covered):** Loops (for/while), Input/Output, Error Handling, Recursion(optional)

### **4. Short Question (Data Types & Primitive Types)**

\* **Question:** Describe the difference between `int` and `double` in Java. Which data type would you use to store a value representing the height of a person in centimeters? Explain your choice.

\* **Expected Response Length:** 3-5 sentences.

\* **(Concept Covered):** Primitive Data Types, Integer vs. Floating-Point

### **5. Long Question (Object-Oriented Programming)**

\* **Question:** Design a simple Java class named `Rectangle` with attributes for width and height. Implement the following methods:

\* `constructor()` to initialize the rectangle's dimensions.

\* `calculateArea()` to calculate and return the area of the rectangle.

\* `toString()` to provide a string representation of the rectangle (e.g., "Rectangle [width=5, height=3]").

Include comments within your code to explain each section. Then, create a `main` method to instantiate an object of the `Rectangle` class and print its area.

\* **Expected Response Length:** Approximately 30-50 Lines of Java Code. (Includes constructor, methods and main)

\* **(Concept Covered):** Classes, Objects, Attributes, Methods, Constructors, Encapsulation, String Representation

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To help me tailor these questions further or provide more relevant assignments, please provide the Faculty Notes, Textbooks, Reference Books, or any other course materials. I can then adjust the difficulty and content to align specifically with your syllabus!